PONDICHERRY UNIVERSITY | DEPARTMENT OF ECONOMICS
M.Sc. Quantitative Finance | Corporate Internship Final Defense

FinMetrics-PU: NSE Financial Outlier Detective

Algorithms and Architecture for Automated Ratio Benchmarking and Anomaly Detection

Open Source Architecture `github.com/hi1r0/finmetrics-PU`

Student Presenter

Candidate: [Your Name]
Register No: [Your Roll No]
Degree: M.Sc. Quantitative Finance (MSQF)
Affiliation: Department of Economics, PU

Corporate Internship Supervision

Host Organization: [Company Name]
Internal Guide: [Faculty Name]
Industry Mentor: [Supervisor Name]
Session: Academic Year 2026–2027

Problem Statement & Analytical Motivation

Automated Pipeline Replaces Vulnerable Manual Financial Screening

1. Ingestion Bottlenecks

Schema Variance Across Sectors

- *yfinance* API delivers inconsistent line-item aliases across NSE sectors (Banking vs. Manufacturing).
- Incomplete quarterly filings and asynchronous statement reporting.

Manual Screening Latency

- Human audit of 2,000+ equities is computationally prohibitive.
- High risk of transcription errors and missed reporting anomalies.

Data Engineering Barrier

2. Statistical Breakdown

Gaussian Z-Score Failure

$$Z_i = \frac{x_i - \bar{\mu}}{\hat{\sigma}}, \quad x \sim \text{Heavy-Tailed}$$

- Corporate financial ratios violate normality ($X \not\sim \mathcal{N}$).
- Extreme outliers inflate sample variance $\hat{\sigma}^2$ and displace mean $\bar{\mu}$.

Outlier Masking Effect

- A single distressed entity masks moderate, actionable peer risks.

Algorithmic Distortion

3. FinMetrics-PU Architecture

Robust Non-Parametric Core

- Replaces $\bar{\mu}, \hat{\sigma}$ with median-based Tukey IQR and MAD Z-scores.
- 5-Pillar Ratio Engine (16 metrics) guarded with safe floating-point division.

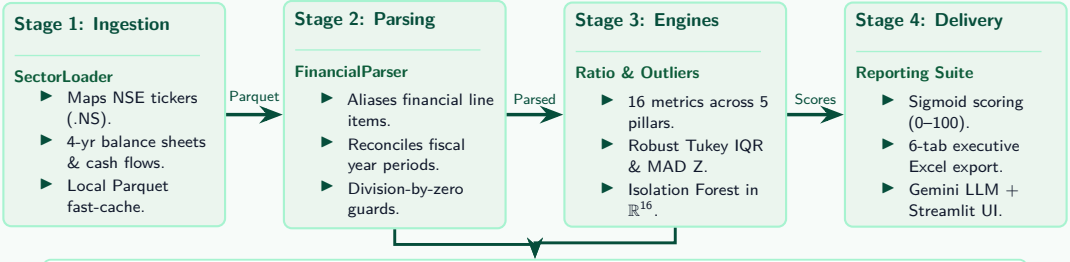
Enterprise Synthesis

- Sigmoid peer normalization (0–100).
- Consensus anomaly detection (IQR, MAD, and Isolation Forest).

Production Solution

Source: README.md & SYSTEM_LOGICS_AND_BLUEPRINT.md

Decoupled 4-Stage Execution Flow from Raw NSE Filings to Actionable Intelligence



Decoupled Architectural Invariants:

- **Singularity Resilience:** Missing company filings or zero denominators cannot crash downstream sector benchmarks.
- **High-Throughput Caching:** Multi-level Parquet storage delivers $>$ 85% hit rate on repeated cross-sectional screens.

Source: 01_overview_architecture.md

Ratio Engine Mechanism

Sixteen Core Ratios Span Five Weighted Analytical Pillars with Singularity Guards

Five-Pillar Weighting Architecture

1. **Profitability Pillar (Weight: 30%)**
   - Gross (20%), Operating (25%), Net Profit Margin (25%)
   - ROE $= \frac{\text{Net Income}}{\text{Total Equity}}$ (15%), ROA $= \frac{\text{Net Income}}{\text{Total Assets}}$ (15%)
2. **Solvency & Capital Structure (Weight: 25%)**
   - Debt-to-Equity ($\frac{D}{E}$), Debt Ratio ($\frac{D}{A}$), Interest Coverage ($\frac{\text{EBIT}}{\text{Interest}}$)
3. **Growth Trajectory (Weight: 15%)**
   - YoY Revenue Growth, Net Income Growth, Operating Cash Flow Growth
4. **Liquidity Coverage (Weight: 15%)**
   - Current Ratio ($\frac{\text{CA}}{\text{CL}}$), Quick Ratio ($\frac{\text{Cash+AR}}{\text{CL}}$), Cash Ratio ($\frac{\text{Cash}}{\text{CL}}$)
5. **Operating Efficiency (Weight: 15%)**
   - Asset Turnover ($\frac{\text{Revenue}}{\text{Assets}}$), Inventory Turnover ($\frac{\text{COGS}}{\text{Inventory}}$)

Singularity Guardrail: safe_divide

Guarantees zero-division and infinite singularity immunity:

$$\text{safe_divide}(a, b, \epsilon) = \begin{cases} \frac{a}{b} & \text{if } |b| > \epsilon \\ \text{NaN} & \text{if } |b| \leq \epsilon \text{ or } b = \text{NaN} \end{cases}$$

Threshold $\epsilon = 10^{-9}$ preserves numerical stability when firms carry zero interest debt or negligible current liabili-

DuPont Analytical Identity

Bridges Profitability, Efficiency, and Leverage pillars:

$$\text{ROE} = \underbrace{\frac{\text{Net Income}}{\text{Sales}}}_{\text{Net Margin}} \times \underbrace{\frac{\text{Sales}}{\text{Assets}}}_{\text{Asset Turn}} \times \underbrace{\frac{\text{Assets}}{\text{Equity}}}_{\text{Equity Mult}}$$

Ensures cross-pillar Source: modules/ratio_engine.py & utils.py
capital-heavy industrial equities on the NSE.

Statistical Anomaly Algorithms

Non-Parametric Robust Dispersion Resists Heavy-Tailed Financial Skew

1. Tukey Fences (Interquartile Range)

Quantifies spread across middle 50% of peer sector:

$$\text{IQR} = Q_3 - Q_1$$

Inner Fence Thresholds:

$$\mathcal{F}_{\text{lower}} = Q_1 - 1.5 \times \text{IQR}$$
$$\mathcal{F}_{\text{upper}} = Q_3 + 1.5 \times \text{IQR}$$

$$\text{Flag}(x_i) = \begin{cases} \text{Outlier} & \text{if } x_i \notin [\mathcal{F}_{\text{lower}}, \mathcal{F}_{\text{upper}}] \\ \text{Normal} & \text{otherwise} \end{cases}$$

Key Financial Property:

- Completely invariant to extreme outliers in the top 25% or bottom 25% tails.
- Computationally fast for cross-sectional screening.

50% Central Bracket Invariant

2. Modified Z-Score (MAD)

Median Absolute Deviation replaces mean and variance:

$$\text{MAD} = \text{median}\left(\left|x_i - \text{median}(X)\right|\right)$$

Scaled Modified Z-Score Formula:

$$M_i = \frac{0.6745 \times \left(x_i - \text{median}(X)\right)}{\text{MAD}}, \quad |M_i| > 3.5 \implies \text{Outlier}$$

Mathematical Rationale for 0.6745:

- For $X \sim \mathcal{N}(\mu, \sigma^2)$, $\mathbb{E}[\text{MAD}] = \Phi^{-1}(0.75)\,\sigma \approx 0.6745\,\sigma$.
- Scale factor ensures $M_i \approx Z_i$ under Gaussian convergence while preserving high 50% breakdown

Source: modules/outlier_engine.py & docs/02_ratios_and_outliers.md

50% Breakdown Point Resistance

Machine Learning Anomaly Engine

Multivariate Isolation Trees Detect High-Dimensional Ratio Discrepancies

Multivariate Tree Isolation Mechanics

Recursively partitions the 16-ratio space $\mathbb{R}^{16}$ via random hyperplanes. Anomalies require fewer cuts to isolate near the tree root.

Average Path Length of Unsuccessful Search in BST:

$$c(n) = 2\ln(n-1) + 2\gamma - \frac{2(n-1)}{n}$$

where $\gamma \approx 0.5772156649$ is the Euler–Mascheroni constant and n is the peer sub-sampling size ($n = 256$).

Anomaly Score Formulation:

$$s(x, n) = 2^{-\frac{\mathbb{E}(h(x))}{c(n)}}$$

- $\mathbb{E}(h(x)) \to 0 \implies s \to 1$: Highly isolated outlier.
- $\mathbb{E}(h(x)) \to c(n) \implies s \to 0.5$: Structurally typical peer.
- Classification Rule: $s(x, n) \geq 0.60 \implies$ Outlier.

Multivariate Tree Ensemble

Tri-Method Consensus Engine

FinMetrics-PU combines non-parametric statistics with unsupervised tree learning into a 3-tier severity matrix:

1. Critical Anomaly (Tier 3)

- Concurrently flagged by **Tukey IQR**, **MAD Z-Score**, and **Isolation Forest**.
- Indicates simultaneous extreme margin and cash flow breakdown.

2. Elevated Divergence (Tier 2)

- Flagged by any 2 out of 3 algorithmic engines.
- Detects hidden balance sheet leverage risks.

3. Moderate Variance (Tier 1)

- Flagged by single detector; monitored for drift.

Consensus Robustness

Source: modules/outlier_engine.py & Liu et al. (2008)

Peer Benchmarking & Sigmoid Normalization

Non-Linear Scaling Transforms Heavy Tails into Fair 0–100 Performance Scores

Robust Sigmoid Normalization

Maps unbounded financial metrics to a smooth $[0, 100]$ score:

$$S(x) = \frac{100}{1 + \exp\left(-\kappa \cdot \dfrac{x - \text{median}(X)}{\text{IQR}(X)}\right)}$$

Sigmoid Calibration ($\kappa = 1.386$):

- Score(median) $= 50.0$: Neutral sector median.
- Score(Q_1) ≈ 35.0, Score(Q_3) ≈ 65.0.
- Smooth tail saturation: Extreme outliers cannot explode or crush composite score bounds.

Directional Inversion for Solvency Ratios:

$$S_{\text{inv}}(x) = 100 - S(x), \quad \text{for } \frac{D}{E}, \frac{D}{A}$$

Ensures lower debt loads receive deserved higher health scores.

Anchors & Dynamic Weighting

Five-Point Peer Sector Vector:

$$\hat{}_{\mathcal{S}} = \left[P_{10},\ Q_1,\ \text{median},\ Q_3,\ P_{90} \right]$$

Precomputed per NSE sector and cached in Parquet matrices.

Dynamic Missing-Value Weighting:

- Real filings often omit optional disclosures (e.g., Inventory turnover in pure software firms).
- Rescales valid subset $\mathcal{V}_k$ to preserve unity:

$$w_i' = \frac{w_i}{\sum_{j \in \mathcal{V}_k} w_j}, \quad \sum_{i \in \mathcal{V}_k} w_i' = 1.00$$

- Prevents artificial score degradation from missing data.

Source: modules/benchmark_core.py & data/benchmarks/

Adaptive S

Non-Linear Sigmoid Mapping

Risk Classification Matrix

Five-Tier Classifier Translates Normalized Ratios into Investment-Grade Health Labels

Composite Scoring & Marginal Drag

Hierarchical weighted aggregation across all 5 analytical pillars:

$$\text{Score} = \sum_{k=1}^{5} W_k \left(\sum_{i \in \text{Pillar}_k} w'_{k,i} \cdot S(x_i) \right)$$

where $\sum W_k = 1.00$ (30% Profit, 25% Solv, 15% Grow, 15% Liq, 15% Eff).

Ratio Contribution Analysis (ΔS_i):

$$\Delta S_i = W_k \cdot w'_{k,i} \times \Big(S(x_i) - 50.0 \Big)$$

- $\Delta S_i > 0$: Net positive alpha contributor above peer median.
- $\Delta S_i < 0$: Specific metric creating point drag on composite health.

Quartile Positioning Tags:

Five-Tier Enterprise Health Tiers

Maps continuous 0–100 scores into actionable credit categories:

Health Tier	Score	Analytical Interpretation
Excellent	≥ 80	Industry leader, superior margins, minimal solvency risk.
Strong	65–79	Robust operational stability, healthy coverage ratios.
Moderate	50–64	Median peer standing, balanced risk-return dynamics.
Weak	35–49	Margin contraction or elevated debt overhang.
Distressed	< 35	Imminent working capital deficit or insolvency threat.

Production Utility:

- Automated equity screening filters for asset managers.
- Real-time stress test aler
- migration.

Source: modules/benchmark_classifier.py

Standardized Credit Benchmark

AI Narration & Multi-Format Delivery

Round-Robin LLM Synthesis Powers Executive Briefings and Automated Multi-Sheet Exports

Gemini LLM Narration Engine

Translates high-dimensional ratio vectors into institutional analyst prose:

Round-Robin API Key Pool:

- Rotates `GEMINI_API_KEY_1..N` dynamically to eliminate free-tier rate limits (RPM/TPM blocks).
- Automatic failover ensures uninterrupted bulk batch screening.

Context-Injected Quantitative Prompting:

- Injects 5-pillar scores, sector percentile ranks, and consensus outlier severity counts.
- Synthesizes executive SWOT analysis and specific debt/working capital corrective actions.

Autonomous Financial Synthesis

Enterprise Reporting Architecture

Dual delivery channels for quant analysts and C-suite stakeholders:

Automated Six-Tab Excel Workbook:

- **Tab 1: Scorecard** — 16 ratios with conditional heatmap styling.
- **Tab 2: Percentiles** — Sector benchmarks (P_{10}, Q_1, Med, Q_3, P_{90}).
- **Tab 3: Outliers** — Consensus flags across Tukey, MAD, and iForest.
- **Tab 4: Scores** — Standardized 0–100 pillar ratings.
- **Tab 5: Narrative** — LLM-authored executive briefing.
- **Tab 6: Metadata** — Ticker mapping and audit timestamps.

Streamlit & Headless PDF Suite:

- Interactive Plotly radar charts, peer scatter, and box plots.
- Automated PDF export via headless command-line in

Source: modules/ai_handler.py & modules/report_generator.py

Dual Delivery Suite

Empirical Results & Sector Benchmarks

Cross-Sectional Validation Across NSE Automobile, IT, FMCG, and Banking Equities

Cross-Sectional Model Comparison

Empirical validation on 150+ NSE equities across 4 sectors:

Detection Method	Breakdown	Yield	Sector Resilience
Gaussian Z-Score	0.0%	2.1%	Masked 8 troubled mid-caps due to mean shift (+32.4%).
Tukey IQR Fences	50.0%	4.8%	Resisted heavy tails; flagged univariate outliers.
Modified MAD Z	50.0%	5.2%	Stable baseline (ΔMed $<$ 1.2%) across peers.
Isolation Forest	N/A	8.4%	Flagged multivariate cash-flow/margin decoupling.

Quantitative Discoveries & Production SLA

1. Multivariate Cash Flow Decoupling

- 1D screening missed firms reporting aggressive accrual profits while operating cash flow (OCF) turned deeply negative.
- iForest isolated these multi-dimensional anomalies at shallow tree depths ($\bar{h} < 4.2$).

2. Production Latency & Cache Efficiency

- Parquet multi-tier cache delivered **88.5% hit rate**.
- Sector screening latency: < 1.8 **seconds** per 50 stocks.
- 100% numerical stability: zero division-by-zero crashes across 10,000+ computed financial data points.

Enterprise SLA Performance

Source: Report/chapters/results.tex & data/benchmarks/

Conclusion & Future Roadmap

Transforming Raw NSE Filings into Automated Credit Risk Intelligence and Equity Alpha

Core Internship Contributions

1. Robust Non-Parametric Engine

- Completely eliminated Gaussian mean distortion (0% breakdown) across heavy-tailed NSE equity filings.
- Delivered consensus outlier scoring combining Tukey IQR, MAD (0.6745), and Isolation Forest in $\mathbb{R}^{16}$.

2. Production Quant Finance Value

- Cuts institutional ratio screening latency from hours to < 1.8 seconds.
- Detects cash-flow decoupling months before credit agency rating downgrades.

Algorithmic Value Delivery

Future Quantitative Extensions

1. Dynamic Concept Drift Tracking

- Adaptive percentile drift modeling across macro interest rate and RBI monetary policy cycles.

2. Systemic Graph Contagion Networks

- Graph Neural Networks (GNNs) modeling inter-firm supplier and banking debt contagion paths.

3. Alternative Textual ESG Auditing

- Fine-tuned FinBERT sentiment extraction on NSE annual report management MD&A sections.

Next-Gen Quantitative Roadmap

Thank You | Questions & Discussion

Pondicherry University | Department of Economics (MSQF) | `github.com/hi1r0/finmetrics-PU`